GAME DESIGN DOCUMENT · FINAL DRAFT (ASSIGNMENT #2)

Rogue Rigs

A top-down roguelite auto-shooter. You are an independent miner stealing ore from a corporate claim before extraction begins, and the arena itself is your best weapon.

> *They filed this claim last week. We're here first. — the Rigs*

GENRE	ENGINE	RUNTIME AI
Roguelite auto-shooter, manual aim	**Unreal Engine 5 · C++**	**None. AI is a dev-time tool only**
PLATFORM	A RUN	BUILD TARGET
PC (Windows) · single-player	**4 levels, upgrades carry over**	**Playable prototype mid-October**

Decision status: `Approved` decided `Provisional` working direction `Open` not decided yet `Deferred` after the capstone

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§1 Executive summary

World. A mining corporation, the Kessler-Voss Mining Consortium, claims rich asteroid deposits and guards them with automated security robots before its extraction crews arrive. Independent miners, the Rigs, land in that window to strip the ore first. The robots are doing their job: nobody on site is human.

The player's side. One miner on the ground, backed by a small floating drone remote-controlled by a colleague in orbit. The colleague talks through the drone, which is where all dialogue comes from.

Audience. Players of auto-shooters and roguelite arcade games, casual to mid-core, playing solo on PC.



Target look, generated during pre-production (ChatGPT, prompts written from this document). It is a readability reference, not a promise of final fidelity.

§2 Design pillars Approved

Every feature must pass at least one pillar test. A feature that passes none is cut.

P1

Gamefeel / juice

Every shot, impact and destruction feels satisfying in the moment.

Passes if the feature adds feedback the player feels, not just a number.

P2

Manual aim + auto-fire

Weapons fire on their own. The player picks the target, and where and when turrets land.

Passes if the player decides where it happens.

P3

The arena is a weapon

Destructible rock blocks shots, shapes chokepoints and hides ore. Breaking it is a decision.

Passes if the terrain changes how a fight plays out.

§3 Game mechanics

A run Approved

- A run is 4 levels played back to back, with nothing in between. Upgrades earned in a level carry into the next.
- **Win:** reach the extraction shuttle at the end of level 4.
- **Lose:** the miner dies, or misses the shuttle, on any level.

A level, step by step Approved

1 · Land	2 · Survive and mine	3 · Objective	4 · Extract
The corporate alert bar starts filling over time. Every robot killed makes it fill faster.	Robot waves and chest spawns punctuate the level. XP cubes drop from kills.	Default: hack the claim beacon. The drone hacks only while the miner stays in the zone.	The shuttle lands far from the miner, with a countdown before corporate reinforcements arrive.

- **Hacking.** If the miner leaves the zone, the drone follows him and hacking stops. Progress then drains slowly, and the beacon shows its gauge so the player sees why.
- **Risk window.** Between the shuttle landing and the countdown ending, the player chooses: keep collecting ore and XP, or run for it.
- **Objective variants** Provisional: a boss (the heavy robot), or find X special ore (a rare vein, distinct from regular ore). Which objective each of the 4 levels uses Open.

Controls and weapons Approved

- The miner moves freely and aims with the mouse or right stick. Every equipped weapon fires automatically toward the aim, each at its own fire rate.
- **Burst rifle:** fires in bursts; the default weapon.
- **Shotgun:** short range, pushes enemies back.
- **Turrets:** up to 2 at once, placed on the aimed spot with a button, fixed lifetime, cannot be picked up. A turret locks the nearest enemy until it dies, leaves its radius or breaks line of sight, then retargets. Turrets are invincible, untargetable, and have no collision with enemies or projectiles, so they never plug a chokepoint.
- **All projectiles stop on rock blocks**, the player's and the robots'. Ranged robots fire only with line of sight.
- **Damage numbers** Provisional: white for normal hits, larger white-gold for critical hits; hits on the same robot within a short window add up into one number; can be turned off.

The drone Approved

A small drone with a cutting laser, remote-controlled by the colleague in orbit. It has three states:

- **Orbit:** circles the miner by default.
- **Dig:** when the miner stands against a block, the drone cuts that block, right next to him.
- **Hack:** inside an objective zone, it links to the beacon.

It cannot dig and hack at the same time. Every block the player digs costs objective progress, which is the central trade-off of a level.

Terrain and destruction Approved

- The arena is a grid. Rock and ore are placed, and destroyed, one block at a time. The arena edge is a cliff into open space.

- **Shots never break blocks.** Only three things do: exploding barrels, exploding kamikaze robots, and the drone.
- **Ore blocks** are tougher and always drop ore, whatever destroyed them. One kamikaze explosion breaks an ore block, so luring a kamikaze onto a vein is a valid tactic.
- **Explosions** have two damage values: one against terrain (tuned to break an ore block in one hit), one against characters (never a one-shot on the player).
- **Kamikazes** do not target defenses. They charge the player and explode on contact or death, which regularly opens chokepoints: no position holds forever. The share of kamikazes per wave controls how long a chokepoint lasts.

Enemies

ROBOT	BEHAVIOR	READ AT A GLANCE	STATUS
Chaser	Runs at the player, melee	Small, bright company yellow	Approved
Kamikaze	Runs at the player, explodes on proximity or death (same radius and damage both ways), hurting robots, player and terrain	Dark gunmetal, pulsing orange core	Approved
Shooter	Keeps distance, needs line of sight, repositions when blocked, shows a red targeting laser before each shot	Acid green, arm cannon from the elbow	Approved
Heavy	Slow, high HP; boss objective	Steel blue, yellow and black hazard stripes, twice the size	Provisional



Enemy line-up. All share red eyes and a yellow corporate logo plate.

Upgrades Approved

- Collecting XP fills a level-up bar. Each level-up offers 3 upgrades; the player picks 1.
- An upgrade applies to one weapon, or to all weapons at a lower value.
- **Stats:** fire rate, damage, critical chance, critical damage.
- **Effects in scope:** piercing projectiles; 4-direction fire (the main shot stays on the aim, the 3 extra shots are weaker, to protect pillar 2). An EMP effect that stuns machines Provisional.

- **Rarity tiers:** grey, green, blue, purple, yellow. These colors appear only in the interface (upgrade cards, HUD), never on an object in the arena, because the arena reserves them (see §4).

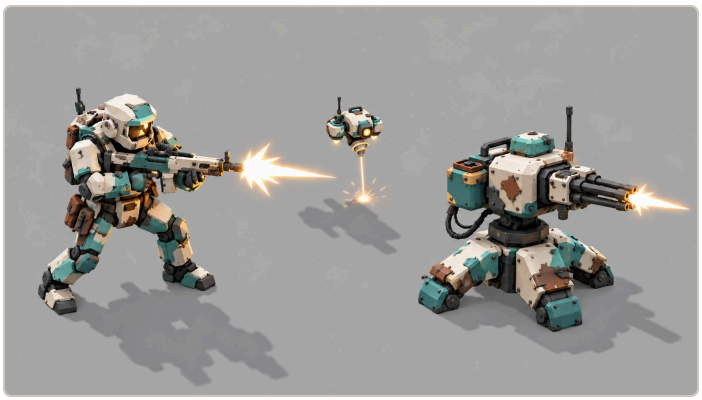
Ore economy

- Ore feeds meta-progression between runs `Deferred`.
- Ore carried during a level is only secured by a successful extraction. Missing the shuttle loses what the miner carries `Approved`. This gives the risk window its stake even before the meta shop exists.
- Losing part of the secured ore when a run fails `Open`: a balance parameter.

§4 Visual direction and readability `Approved`

- **Style:** flat-shaded low-poly. Camera: steep, near-vertical perspective, the miner always centered. Blocks are always low, so nothing hides a unit. Clean floor, no scattered pebbles.
- **Readability rule:** terrain separates by brightness (breakable rock is lighter; ore glows), cracks always mean "breakable", and enemies use colors the terrain never uses.

ELEMENT	COLOR
Breakable rock	Light grey tinted violet, visible cracks
Ore rock	Same base with glowing violet crystals
XP and resource pickups	Glowing cyan
Miner, turrets, drone	White, teal and rust, patched salvage
Player shots, mining laser	White-gold
Enemy shots, targeting laser, barrel	Red
Enemies	Yellow, gunmetal, acid green, steel blue (reserved for enemies in every biome)



The Rigs' side: everything repaired and bolted together, in contrast with the clean corporate robots.

§5 AI architecture

No AI runs while you play. Every agent works during development. Level generation at runtime is a deterministic C++ algorithm that the agents helped design.

Orchestration. I give objectives to a single Orchestrator, acting as lead architect. It splits them into tasks and delegates each to one specialized agent. Agents do not talk to each other directly: they read their inputs from, and write their outputs to, the project vault (tasks, this GDD split into agent-sized files, technical docs). The Orchestrator reads the vault through its index, never whole.

AGENT	PRODUCES	OUTPUT FORMAT	TOKEN CEILING / TASK
Orchestrator	Task breakdown; detects GDD features missing from the code	**`tasks/<id>.md`**: goal, agent, inputs, expected output, ceiling, status	20k
C++ agent	Gameplay, enemy AI, grid, generator, logs; researches the codebase itself	**`.h/.cpp`** in **`Source/RogueRigs/`**, Unreal conventions, must compile, plus a change summary	80k
BP agent	Blueprints derived from C++ classes, UMG widgets, Niagara hookups	Unreal assets created through Unreal MCP	40k
Level-Design agent	Fixed prototype layouts, then generator rules	Grid JSON **`{"cells": [{"x","y","type"}]}`**, types: rock, ore, special_ore, barrel, spawn, beacon, exit; rules JSON (densities, minimum distances)	30k
Balance agent **`Ephemeral`**	Tuned values, only once playtest logs exist	CSV rows for Unreal DataTables (weapons, robots, waves, rarities) + a note citing the log metrics used	30k
Text agent	Colleague and miner barks, upgrade texts	JSON: **`line_id`**, **`speaker_id`**, **`trigger`**, **`tone`** (fixed list of 5), **`text`** of 8 words max	15k
Style Guide agent	Checks text tone and image prompts against the style guide; the Evaluator in a Generator → Evaluator → Refiner loop	JSON per item: **`item_id`**, **`verdict`**, **`rule`**, **`reason`**, **`suggestion`**	15k
Art agent	Image prompts and 3D source images	One **`.md`** prompt per asset + reference images	15k
Animation agent	Procedural animation for the rigid-part robots; retargeting for the miner	C++ AnimInstance classes + a retarget list	50k

AGENT	PRODUCES	OUTPUT FORMAT	TOKEN CEILING / TASK
Sound agent	Finds openly licensed sounds and scores them against the style guide; music per alert level through my existing Suno pipeline	`sound_manifest.json`: `event_id`, description, source, file, variations	20k
Adversarial QA agent	Plays to break the game	Bug reports in `.md`	`Deferred`

Dialogue tone (text agent + Style Guide): a cocky, odds-dismissing smuggler crossed with a tired, sarcastic blue-collar worker. Combat barks of 3 to 8 words. Self-mockery is fine; mocking the player is not. The jokes target the company. No famous film lines, quoted or paraphrased.

Examples: "Send the invoice to corporate." · "They noticed. Great. Love it when they notice." · "Of course it's on the other side of the rock."

```
{"item_id": "bark_042",
 "verdict": "fail",
 "rule": "tone.no_player_mockery",
 "reason": "Mocks the player, not the
company",
 "suggestion": "Company's gonna love
this invoice."}
```

§6 Technical strategy

- **Why Unreal:** C++ for all gameplay logic (the course's advice, and my background); Niagara for debris and damage numbers at horde scale; the Environment Query System for shooters looking for a firing position; Behavior Trees; instanced meshes for the grid; Unreal MCP so agents can create assets.
- **Code split:** logic in C++. Blueprints stay a thin layer: classes derived from C++ for placement in levels, events that call C++ functions, a few visual effects such as camera shake.
- **Grid rendering:** a block's mesh depends on its open sides. With rotation, 6 shapes cover every case (buried, 1 open side, corner, corridor, wall end, isolated), each with a few variants. Hierarchical Instanced Static Meshes per asset type. A destroyed block plays a particle burst and its neighbors swap to the shape matching their new open sides.
- **Robot deaths:** robots are built from rigid parts, each bound to one bone. On death, those parts are thrown from the last hit's direction as Niagara mesh particles (floor collision only), then shrink and sink. No physics interaction with anyone, and debris systems are pooled.
- **Assets:** Meshy (low-poly mode) for robots, drone and turret; Tripo for the miner (auto-rig). Rock blocks, crystals and barrels are modeled by hand. Every generated mesh is cleaned and decimated.
- **Run logs** (input for the balance agent): one structured file per run, one entry per level. Terrain destroyed by source and type; damage dealt per weapon; kills by robot type and by source (weapon, explosion); upgrades picked (order and frequency); time per level; damage taken by source; XP collected; level outcome (death, extracted, missed shuttle); time left on reaching the shuttle; ore collected during the risk window; events met and their outcome.

Constraints acknowledged

- **Time:** one person, 5 to 6 hours a week, about 6 weeks left.
- **AI budget:** a Claude Code Pro subscription with a monthly quota. There is no money cap to manage, but when the quota runs out, work pauses until it resets. Token ceilings per task (§5) are first estimates, corrected after each milestone retrospective.
- **Performance:** the navigation mesh must rebuild around destroyed blocks during play; line-of-sight checks for shooters and turrets are throttled rather than run every frame.
- **3D assets** are the weakest point of AI generation: expect cleanup on every model.

§7 Scope and milestones Approved

MILESTONE	CONTENT	TARGET
M1 · Playable prototype	One fixed arena from a Level-Design JSON; movement and aim; burst rifle and shotgun; chaser and kamikaze; grid with block destruction (barrel, kamikaze, drone); XP, level-up, stat upgrades; beacon hack and shuttle extraction	Mid-October
M2 · Full run	Shooter (line of sight, EQS, targeting laser); turrets; 4-level run; grid generator; rarities; piercing and 4-direction fire; logs; damage numbers	End of October
M3 · Polish	Heavy robot as boss; sound and music; barks; juice (debris, VFX, critical-hit pop); EMP if time allows	November 9 submission

Cut rule: anything that does not fit in M3 goes to the wishlist, which lives in the vault, outside this document. Already cut: the meta-progression shop, per-weapon-type upgrades, grenades, the adversarial QA agent.

§8 What changed since Assignment #1

Instructor feedback on the draft: "You may need additional agents as you get to the polish phases for animation and sound." Addressed in change 9.

#	CHANGE	WHY
1	Placeholder title and no setting → **Rogue Rigs**, ore theft on a corporate asteroid claim	The draft had no world. The lore now explains each mechanic: alert bar, beacon hack, shuttle, robots
2	Win and loss defined for the whole run (4 levels)	The draft only defined them per level
3	Ore is secured only by extraction	Logic gap: ore rewarded nothing without the meta shop. The risk window now has a stake
4	Terrain on a grid; shots no longer break rock; three destruction sources	Destruction becomes a decision instead of a side effect of shooting, which makes pillar 3 real
5	New: the drone (dig, hack, dialogue)	One tool for two jobs creates the central trade-off of a level

#	CHANGE	WHY
6	Dropped Chaos Destruction for grid autotiling and Niagara debris	Performance with hordes, and a scope one person can build
7	Four-robot roster with a readability color code	The shooter changes how the player moves, so it must be spotted in under a second
8	Endless-wave objective replaced by "find special ore"	The endless wave added nothing over the base loop; ore hunting uses pillar 3
9	Agents now state their output format; added animation, sound, art and style guide agents	Instructor feedback, and agent role clarity
10	Milestones M1 to M3 and a separate wishlist	The core had grown past the 48-hour test; this keeps it buildable in the weeks left

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Property of Kessler-Voss Mining Consortium. Or it was.